

2 Monomials—Laws of Indices—Degree

Exercises II

1. Simplify—

$$\frac{5^7 \times 12^4 \times 32^2 \times (3^2 \times 4^2 \times 5)^2}{(3 \times 15 \times 2^3)^{10}}.$$

First, we simplify the numerator,

$$\begin{aligned} & 5^7 \times 12^4 \times 32^2 \times (3^2 \times 4^2 \times 5)^2 \\ &= 5^7 \times (2 \times 2 \times 3)^4 \times (2^5)^2 \times (3^2 \times (2 \times 2)^2 \times 5)^2 && \text{(arithmetical factorization)} \\ &= 5^7 \times 2^4 \times 2^4 \times 3^4 \times 2^{10} \times 3^4 \times 2^4 \times 2^4 \times 5^2 && ((ab)^n = a^n b^n; (a^m)^n = a^{mn}) \\ &= 5^9 \times 3^8 \times 2^{26} && \text{(commutation; } a^m a^n = a^{m+n}) \end{aligned}$$

Next, we simplify the denominator,

$$\begin{aligned} & (3 \times 15 \times 2^3)^{10} \\ &= 3^{10} \times 15^{10} \times 2^{30} && ((ab)^n = a^n b^n; (a^m)^n = a^{mn}) \\ &= 3^{10} \times (3 \times 5)^{10} \times 2^{30} && \text{(arithmetical factorization)} \\ &= 3^{10} \times 3^{10} \times 5^{10} \times 2^{30} && ((ab)^n = a^n b^n) \\ &= 3^{20} \times 5^{10} \times 2^{30} && (a^m a^n = a^{m+n}) \end{aligned}$$

Finally, we simplify the fraction,

$$\begin{aligned} & \frac{5^9 \times 3^8 \times 2^{26}}{3^{20} \times 5^{10} \times 2^{30}} \\ &= \frac{1}{3^{12} \times 2^4 \times 5} && \left(\frac{a^m}{a^n} = \frac{1}{a^{n-m}}; n > m\right) \\ &= \frac{1}{42515280} \end{aligned}$$

2. Which is greater, $(2^2)^{2^2}$ or 2^{2^2} ? Find the difference between them.

Observe that $2^{2^{2^2}} = 2^{2^4} = 2^{16} = 65536$, whereas $(2^2)^{2^2} = (2^2)^4 = 2^8 = 256$.
Therefore $2^{2^{2^2}}$ is larger.

3. Simplify—

$$\frac{2^{2^2}}{2(2^2)^2}.$$

$$\frac{2^{2^2}}{2(2^2)^2} = \frac{(2 \times 2)^2}{2 \times (2 \times 2)^2} = \frac{1}{2}$$

4. Simplify—

$$\frac{36^2 a^7 b^5 c^3 d^2}{81 a^4 b^3 c^3}.$$

$$\frac{36^2 a^7 b^5 c^3 d^2}{81 a^4 b^3 c^3} = \frac{(9 \times 4)^2 a^7 b^5 c^3 d^2}{(9 \times 9) a^4 b^3 c^3} = \frac{9^2 \times 4^2 \times a^7 b^5 c^3 d^2}{9^2 a^4 b^3 c^3} = 4^2 a^3 b^2 d^2 = 16 a^3 b^2 d^2$$

5. Express in its simplest form—

$$\left(\frac{a^4 c^3 y^2}{a^3 b^2 c^3 x^2 y^2} \right)^2 \times \left(\frac{a^3 b^3 x^3}{a^3 b^2 c^3 x^3 y^2} \right)^3 \times \left(\frac{b^2 c^4 x^2 y}{a^4 b^3 x^3 y^3} \right)^4.$$

$$\begin{aligned}
& \left(\frac{a^4 c^3 y^2}{a^3 b^2 c^3 x^2 y^2} \right)^2 \times \left(\frac{a^3 b^3 x^3}{a^3 b^2 c^3 x^3 y^2} \right)^3 \times \left(\frac{b^2 c^4 x^2 y}{a^4 b^3 x^3 y^3} \right)^4 \\
&= \left(\frac{a}{b^2 x^2} \right)^2 \times \left(\frac{b}{c^3 y^2} \right)^3 \times \left(\frac{b^2 c^4 x^2 y}{a^4 b^3 x^3 y^3} \right)^4 & \left(\frac{a^m}{a^n} = a^{m-n}; m > n \right) \\
&= \left(\frac{a}{b^2 x^2} \right)^2 \times \left(\frac{b}{c^3 y^2} \right)^3 \times \left(\frac{c^4}{a^4 b x y^2} \right)^4 & \left(\frac{a^m}{a^n} = \frac{1}{a^{n-m}}; n > m \right) \\
&= \frac{(a)^2}{(b^2 x^2)^2} \times \frac{(b)^3}{(c^3 y^2)^3} \times \frac{(c^4)^4}{(a^4 b x y^2)^4} & \left(\left(\frac{a}{b} \right)^m = \frac{a^m}{b^m} \right) \\
&= \frac{a^2}{(b^2)^2 (x^2)^2} \times \frac{b^3}{(c^3)^3 (y^2)^3} \times \frac{(c^4)^4}{(a^4)^4 (y^2)^4 b^4 x^4} & ((ab)^m = a^m b^m) \\
&= \frac{a^2}{b^4 x^4} \times \frac{b^3}{c^9 y^6} \times \frac{c^{16}}{a^{16} y^8 b^4 x^4} & ((a^m)^n = a^{mn}) \\
&= \frac{a^2 b^3 c^{16}}{b^4 x^4 c^9 y^6 a^{16} y^8 b^4 x^4} \\
&= \frac{a^2 b^3 c^{16}}{b^8 x^8 c^9 y^{14} a^{16}} & (a^m a^n = a^{m+n}) \\
&= \frac{c^{16}}{b^5 x^8 c^9 y^{14} a^{14}} & \left(\frac{a^m}{a^n} = \frac{1}{a^{n-m}}; n > m \right) \\
&= \frac{c^7}{b^5 x^8 y^{14} a^{14}} & \left(\frac{a^m}{a^n} = a^{m-n}; m > n \right)
\end{aligned}$$

6. Simplify—

$$\begin{aligned}
& \left(\frac{45a^3 b^2 c^4}{27a^2 b^2 c} \right)^2 \times \left(\frac{243a^4 b^4 c^4 x}{180a^2 b c} \right)^2 \\
&= \left(\frac{(9 \times 5)a^3 b^2 c^4}{(9 \times 3)a^2 b^2 c} \right)^2 \times \left(\frac{3^5 a^4 b^4 c^4 x}{(2^2 \times 3^2 \times 5)a^2 b c} \right)^2 \\
&= \left(\frac{5ac^3}{3} \right)^2 \times \left(\frac{3^3 a^2 b^3 c^3 x}{2^2 \times 5} \right)^2 \\
&= \frac{5^2 a^2 c^6}{3^2} \times \frac{3^6 a^4 b^6 c^6 x^2}{2^4 \times 5^2} \\
&= \frac{3^4 a^6 b^6 c^{12} x^2}{2^4} \\
&= \frac{81a^6 b^6 c^{12} x^2}{16}
\end{aligned}$$

7. Simplify—

$$(3xyz)^2 \times \frac{x^6 y^9}{x^9 y^6} \div \left(\frac{3x^2 z^2}{y^2 z^2} \right)^2.$$

8. Simplify—

$$\frac{(x^3 y^2 z^2)^7 \times (y^3 z^2 x^2)^7 \times (z^3 x^2 y^2)^7}{\left(\frac{x^2}{yz} \right)^7 \times \left(\frac{y^2}{zx} \right)^7 \times \left(\frac{z^2}{xy} \right)^7}.$$

9. Simplify—

$$\left(\frac{x^m}{x^n} \right)^{m+n} \times \left(\frac{x^n}{x^l} \right)^{n+l} \times \left(\frac{x^l}{x^m} \right)^{l+m}.$$

10. Simplify—

$$\left\{ \frac{(a^{p-q})^q \times (a^{q-r})^{r-p}}{(a^{r+q})^{r-q}} \right\}^{1/r}.$$

11. Simplify—

$$\frac{(x^{a-b} \times x^{b-c})^a \times \left(\frac{x^a}{x^c} \right)}{(x^a \times x^c)^a \div (x^{a+c})^c}.$$

12. Simplify—

$$\left(\frac{x^p}{x^q} \right)^{p+q} \div \left(\frac{x^{p+q}}{x^{p-q}} \right)^{p^2/q}.$$

13. Simplify—

$$\left\{ \left(\frac{x^l}{x^m} \right)^l \times \left(\frac{x^m}{x^l} \right)^m \right\} \div \{ (x^l)^l \times (x^m)^m \} \times \{ (x^m)^l \times (x^l)^m \}.$$

14. Prove that—

$$\frac{(yz)^{qr}(zx)^{rp}(xy)^{pq}}{(y^{q-1}z^{r-1})^p(z^{r-1}x^{p-1})^q(x^{p-1}y^{q-1})^r} = \frac{(xyz)^{p+q+r}}{x^p y^q z^r}.$$

15. Distribute the product—

$$(x^{b-c} - x^{c-a} + x^{a-b}) \left(\frac{1}{x^{b-c}} + \frac{1}{x^{c-a}} + \frac{1}{x^{a-b}} \right).$$

16. Distribute—

$$\left(a^p + \frac{1}{b^q}\right) \left(a^q + \frac{1}{b^p}\right)^2.$$

17. If $m = a^x$, $n = a^y$, $a^2 = (m^y n^x)^z$; show that $xyz = 1$.